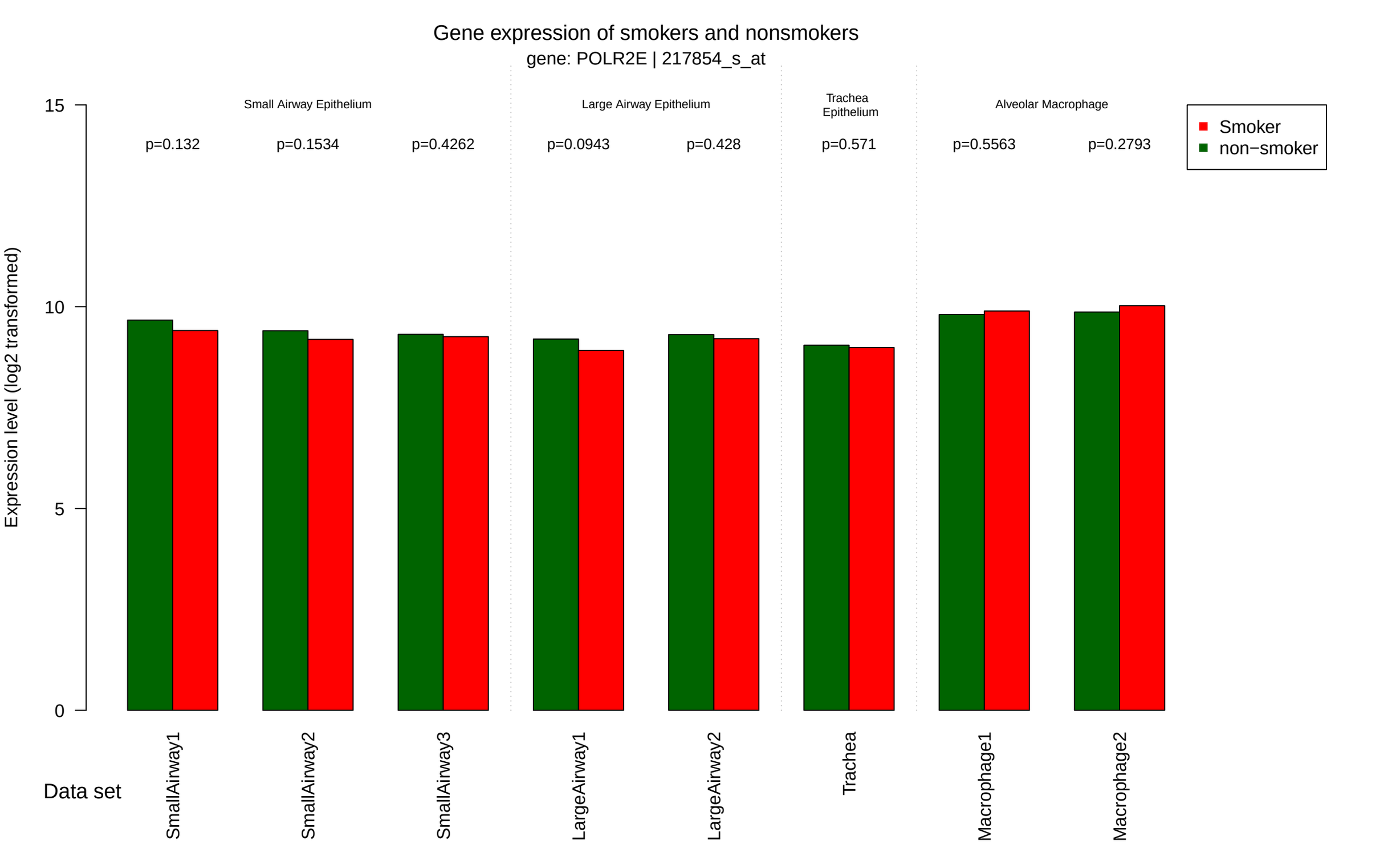
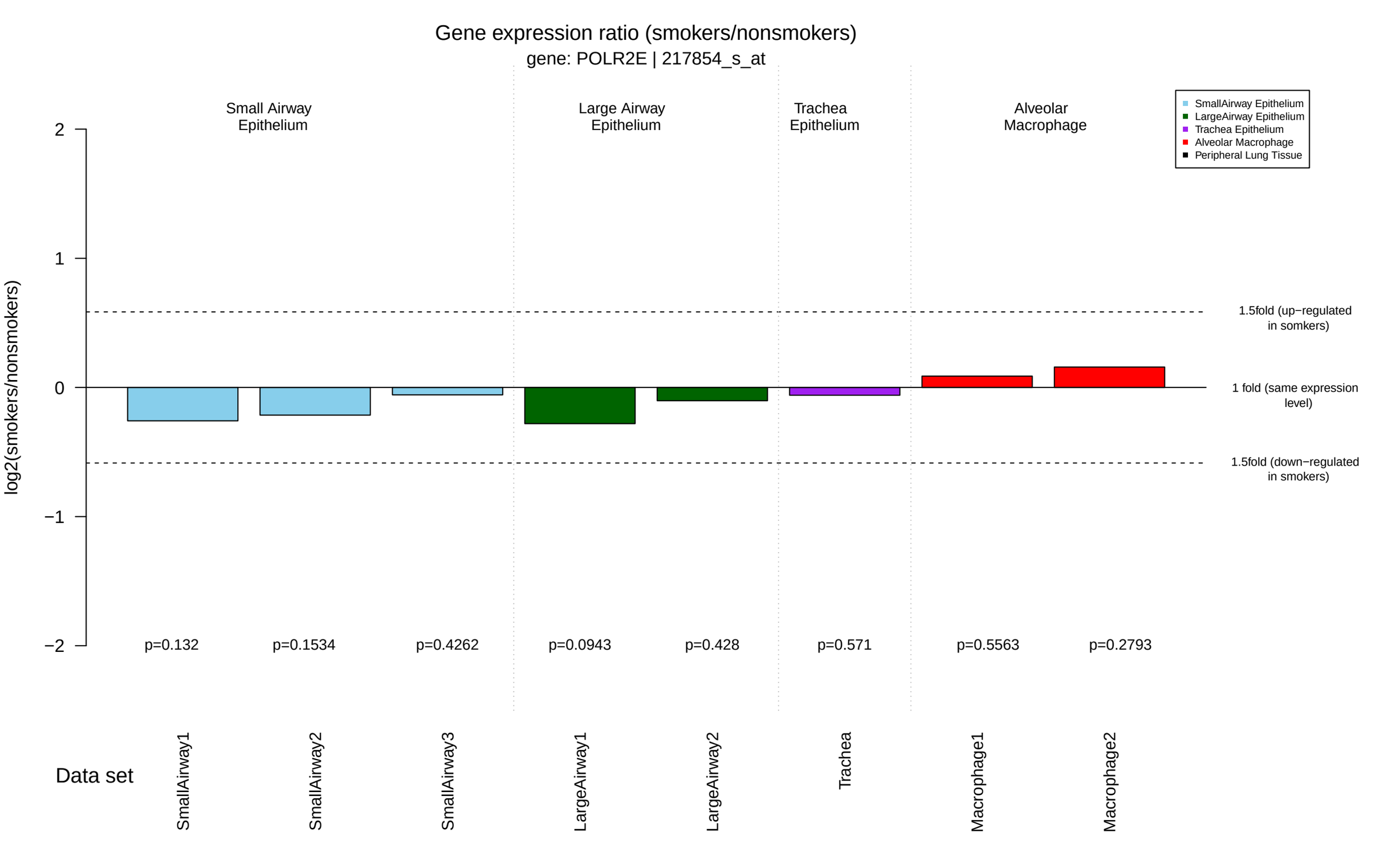
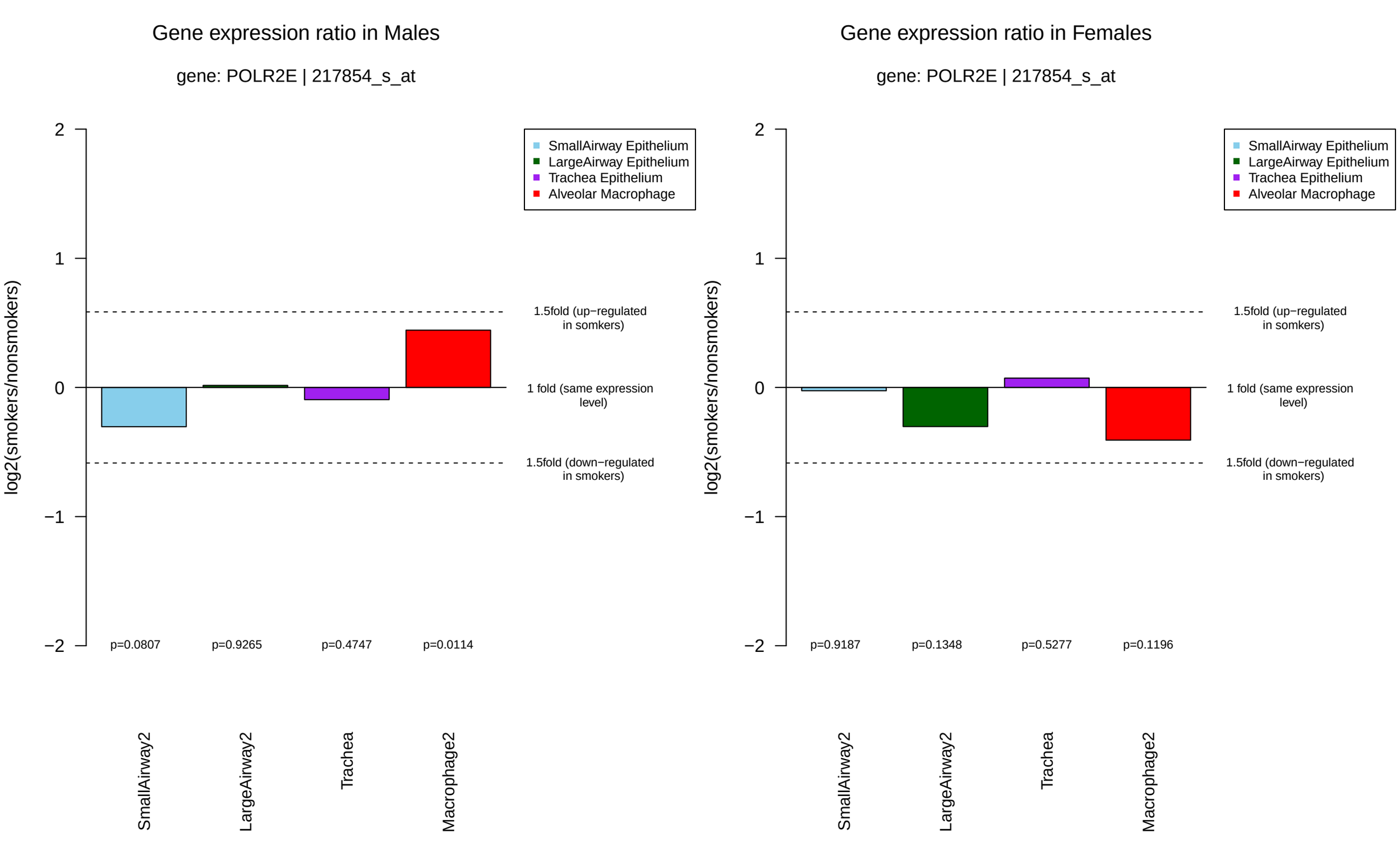
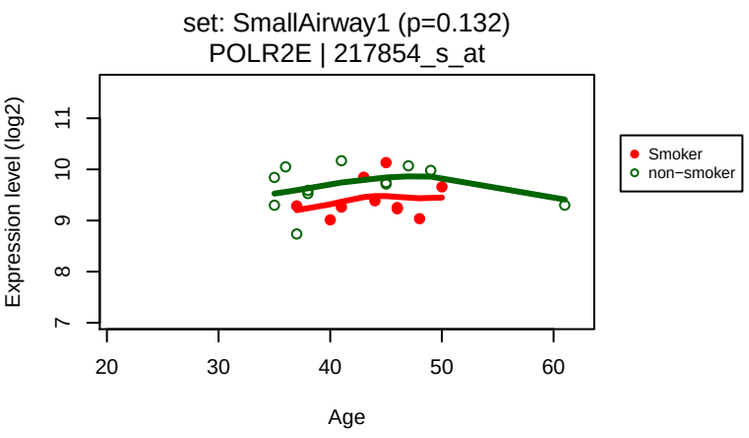
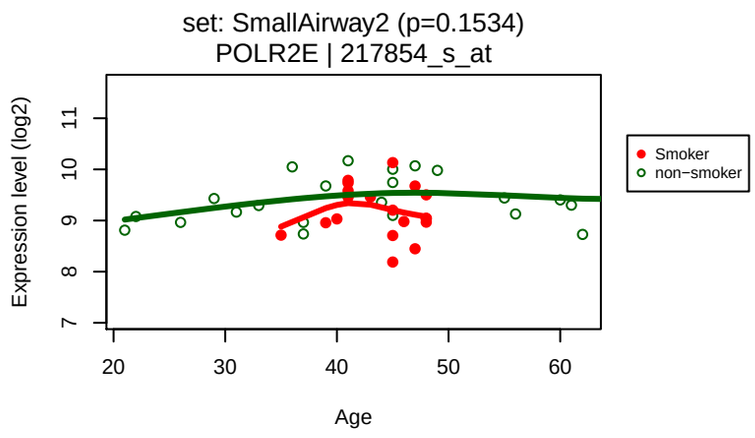
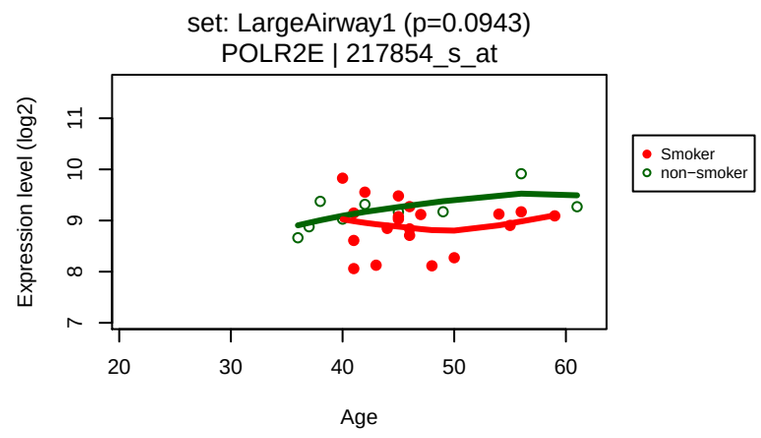

Gene expression of smokers and nonsmokers
gene: POLR2E | 217854_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.132
p=0.1534
p=0.4262
p=0.0943
p=0.428
p=0.571
p=0.5563
p=0.2793
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

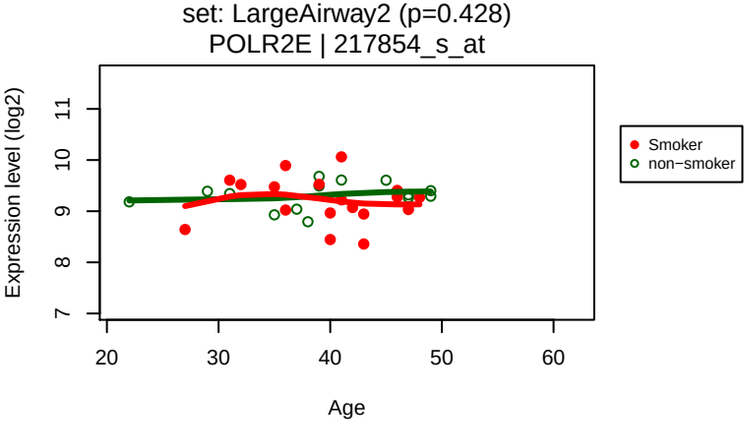

Gene expression ratio (smokers/nonsmokers)
gene: POLR2E | 217854_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.132
p=0.1534
p=0.4262
p=0.0943
p=0.428
p=0.571
p=0.5563
p=0.2793
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

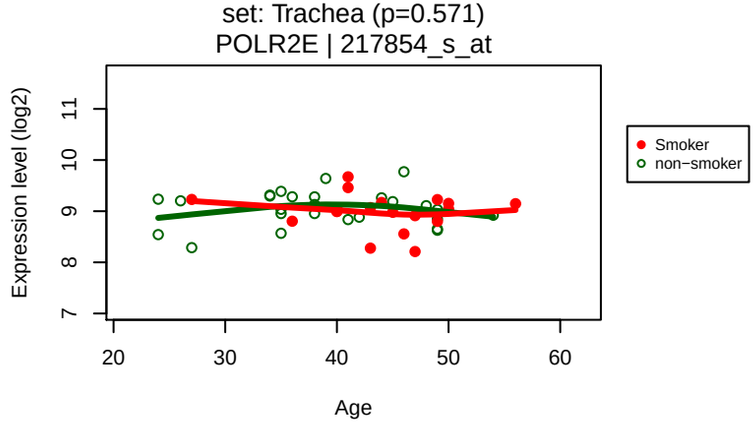

Gene expression ratio in Males
gene: POLR2E | 217854_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0807
p=0.9265
p=0.4747
p=0.0114
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR2E | 217854_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9187
p=0.1348
p=0.5277
p=0.1196
SmallAirway2
LargeAirway2
Trachea
Macrophage2

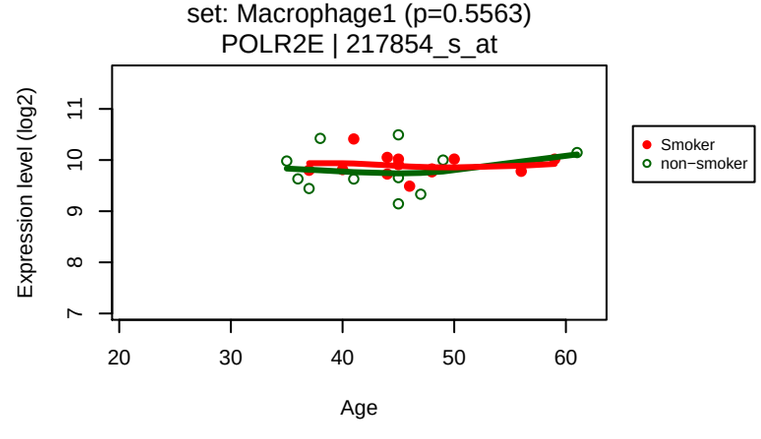

set: SmallAirway1 (p=0.132)
POLR2E | 217854_s_at
Expression level (log2)
Age
Smoker
non−smoker

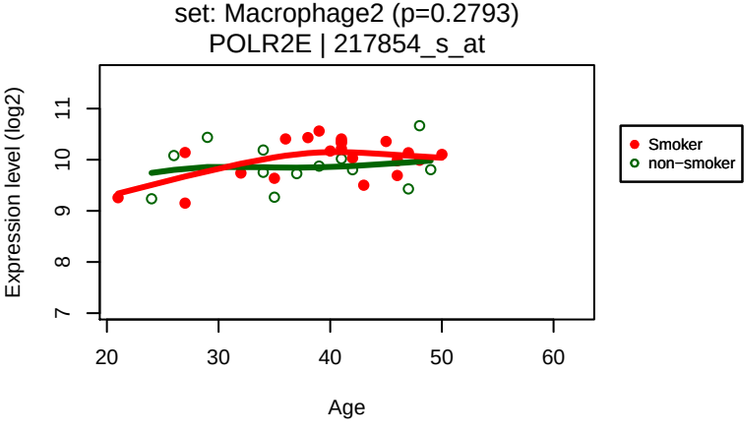

set: SmallAirway2 (p=0.1534)
POLR2E | 217854_s_at
Expression level (log2)
Age
Smoker
non−smoker

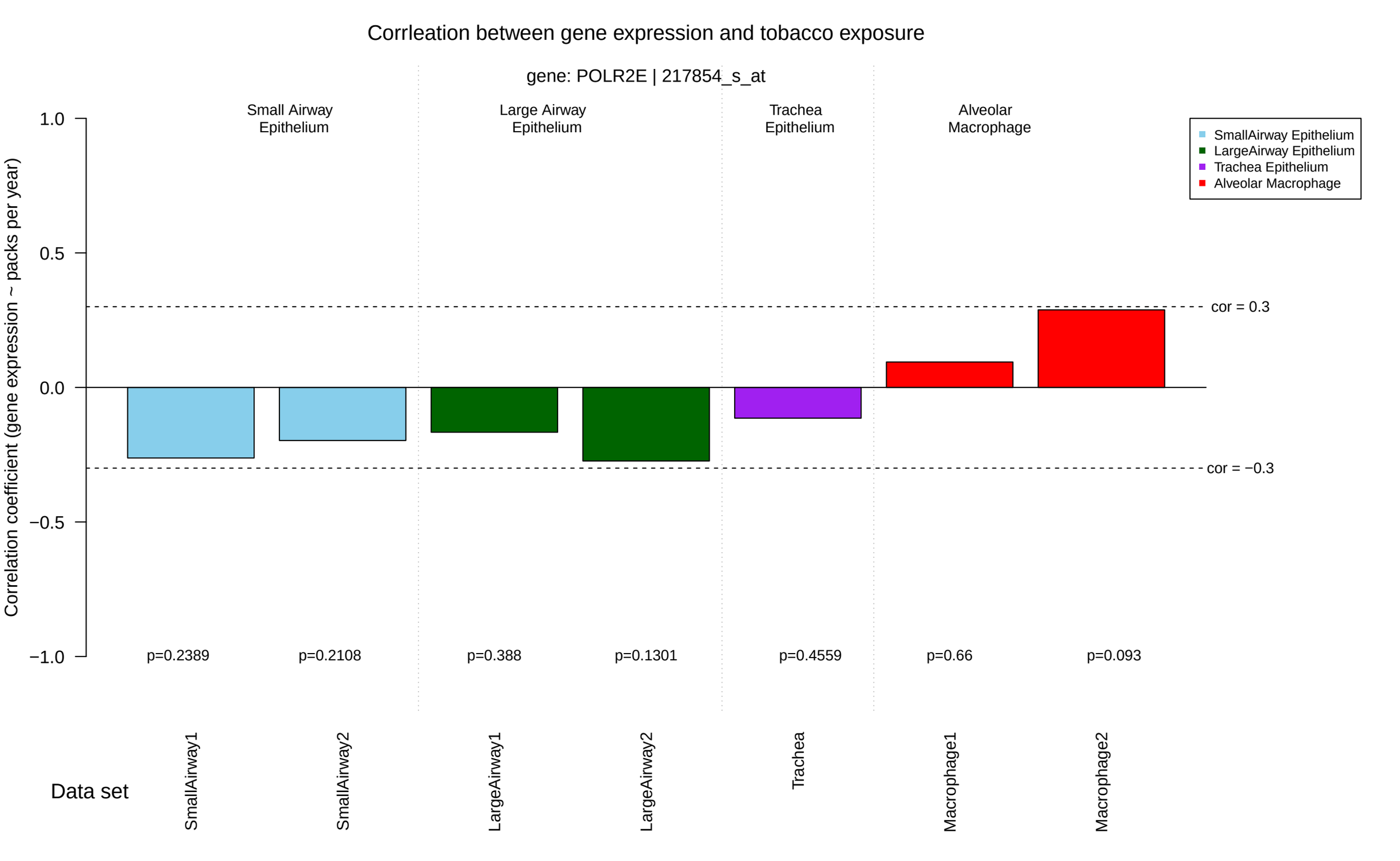

set: LargeAirway1 (p=0.0943)
POLR2E | 217854_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.428)
POLR2E | 217854_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.571)
POLR2E | 217854_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5563)
POLR2E | 217854_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2793)
POLR2E | 217854_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR2E | 217854_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2389
p=0.2108
p=0.388
p=0.1301
p=0.4559
p=0.66
p=0.093
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2